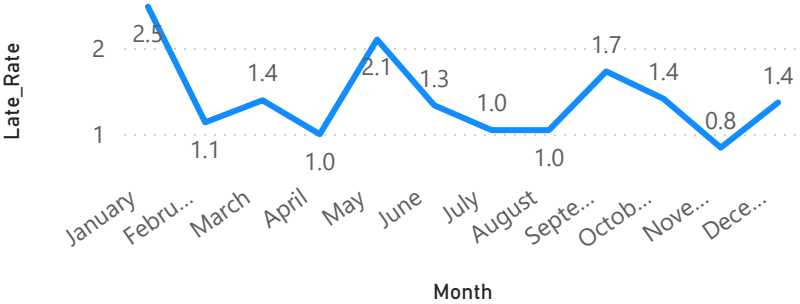


Carrier_Name	Avg_Delivery_Days	Cost_Per_Shipment	OnTime_Rate
V44_3	2.98	10.27	100.00
V444_1	1.50	9.76	99.57
V444_0	2.50	1.81	97.04
Total	1.89	6.75	98.61

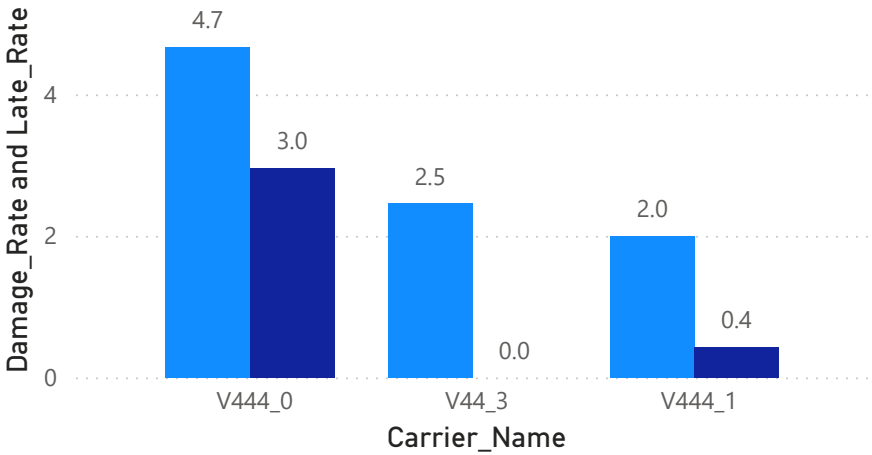
Late_Rate by Month



Insights : Late delivery rate spikes in April-May and October-November, likely seasonal peaks or capacity constraints during those periods.

Damage_Rate and Late_Rate by Carrier_Name

● Damage_Rate ● Late_Rate

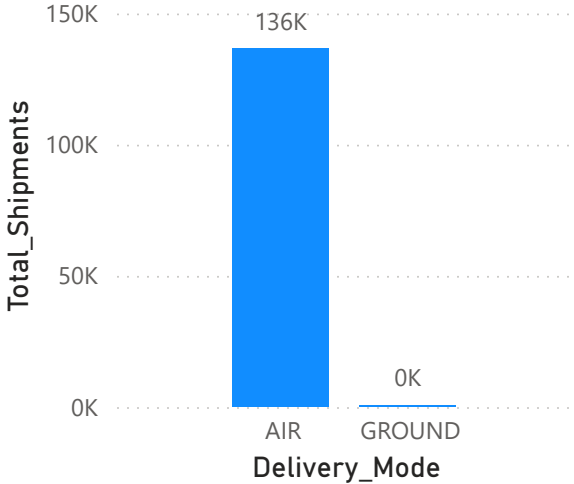


Carrier_Name Delivery_Mode

- ☐ V44 3 ☐ AIR
- ☐ V444 0 ☐ GROUND
- ☐ V444 1

Insights : Carrier V444_0 has the highest damage rate and late rate.
V444_1 has least damage rate and late rate.

Total_Shipments by Delivery_Mode

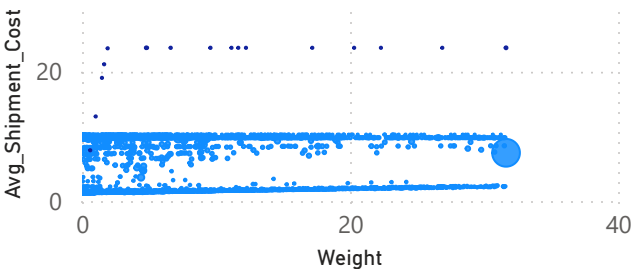


Insights : Air is the most used delivery mode as we can see from this graph.
Air has 136K total shipments where ground has very few total shipment

Insights : Carrier V444_1 has the most reliable carrier as it has least avg delivery days and least late rate and sum of accuracy flag is much higher than other carriers.

Avg_Shipment_Cost and Total_Shipments by Delivery_Mode and Weight

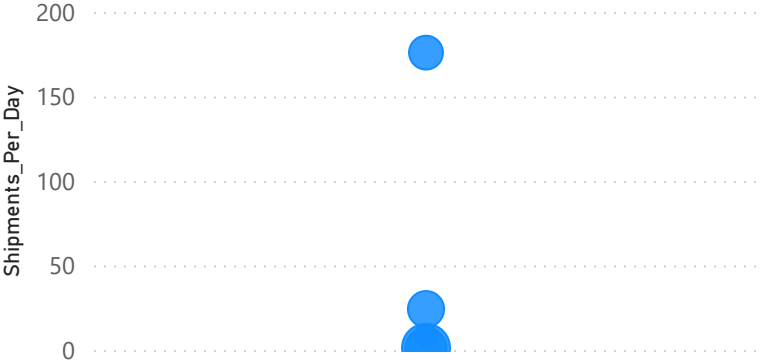
Delivery_M... ● AIR ● GROUND



Insights : Ground mode dominates both volume and cost, confirming it's the primary transport method for this operation.

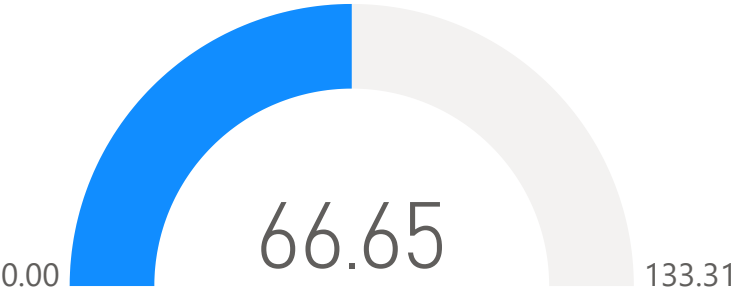
Utilization_Pct	Shipments_Per_Day	Cost_Per_Labour_Hour
66.65	186.80	58.57

Shipments_Per_Day and Cost_Per_Shipment by Warehouse_ID



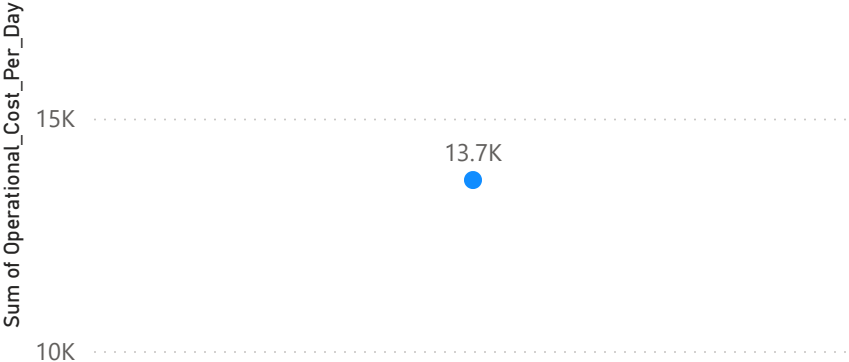
Insights : One warehouse stands out with significantly higher daily shipments but similar cost per shipment to lower-volume peers, indicating superior economies of scale.

Utilization_Pct



Insights : Warehouse operates in the fair-to-good utilization band (below optimal 70-85% range), indicating capacity headroom for additional volume without major infrastructure investment.

Sum of Operational_Cost_Per_Day



Insights : Operational costs vary significantly across warehouses, with a few high-cost outliers driving disproportionate spend.

Warehouse_ID	Shipments_Per_Day	Warehouse_Utilization_%
PLANT03	175.96	71.68
PLANT04	1.00	67.35
PLANT08	8.70	54.04
PLANT09	9.09	61.69
PLANT12	24.19	67.85
PLANT13	8.78	55.02
PLANT16	1.32	51.97
Total	186.80	

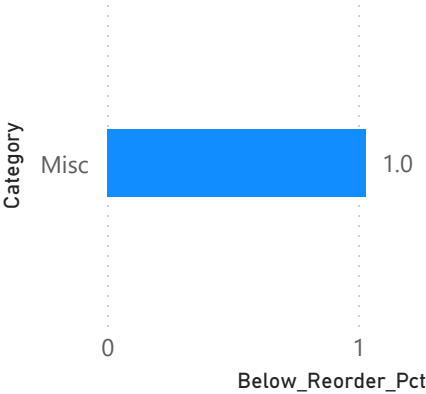
Insights : Plant03 has the highest shipments per day i.e. 175.96 and highest warehouse utilization_% 71.68.
Plant 16 has least warehouse utilization 51.97.

Category	PLANT03	PLANT04	PLANT08	PLANT09
Misc	0.00	0.00	0.00	0.00
Total	0.00	0.00	0.00	0.00



Insights : Here can see average stock out days per warehouse by category.
 We can see 0 stock out days for mic category for all warehouses.

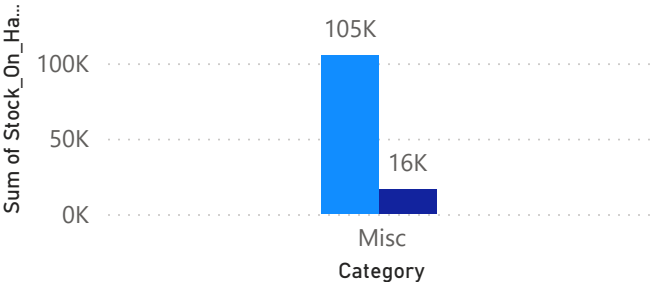
Below_Reorder_Pct by Category



Insights : From this we can see the below reorder percentage.
 We have only 1 category and it has 1.0 reorder percentage.

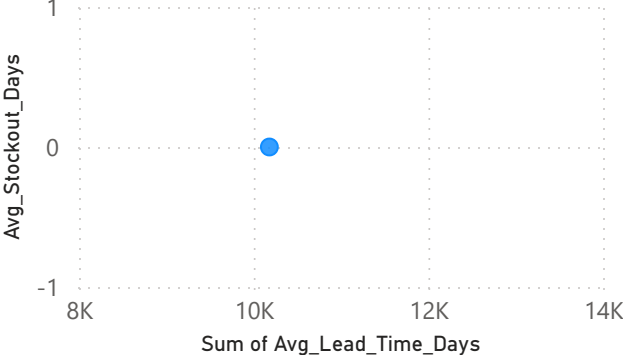
Sum of Stock_On_Hand and Sum of Reorder_Level by Category

● Sum of Stock_On_Hand ● Sum of Reorder_Level



Insights : By this graph we can see the stock on hand and reorder level by Category.
 For category Mic stock on hand is 105K and reorder level is 16K.

Sum of Avg_Lead_Time_Days and Avg_Stockout_Days by Category



Insights : Here we can find the relationship between the avg_lead_time_days and avg stock out days.
 The value we got is 0 so there is no relation

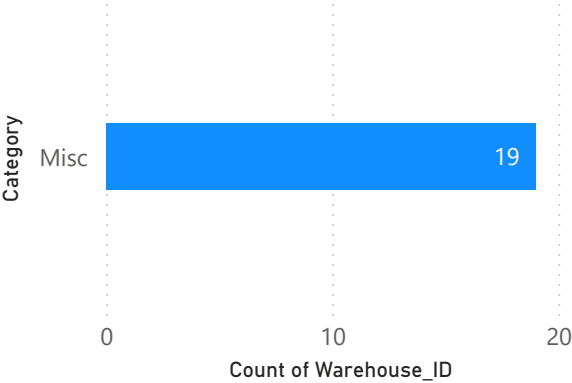
Insights : In this graph we can see the plant wise total carrying cost and stock in hand.
 There is only one category so plant wise numbers we can see for category misc.

Carrying_Cost_Total and Sum of Stock_On_Hand by Warehouse_ID and Category



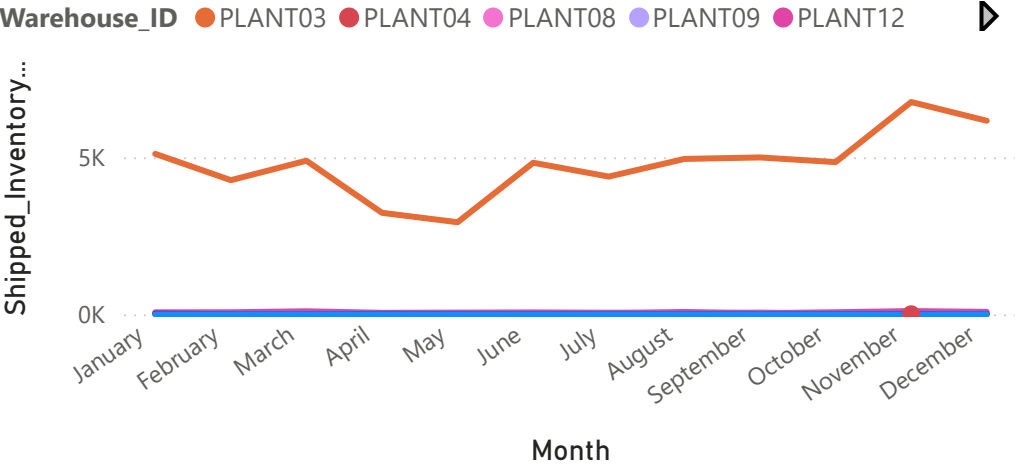
Category	Warehouse_ID	Avg_Stockout_Days	Sum of Carrying_Cost_Per_Unit
Misc	CND9	0.00	0.09
Misc	PLANT01	0.00	24.20
Misc	PLANT02	0.00	11.60
Misc	PLANT03	0.00	78.10
Misc	PLANT04	0.00	12.06
Total		0.00	208.10

Count of Warehouse_ID by Category



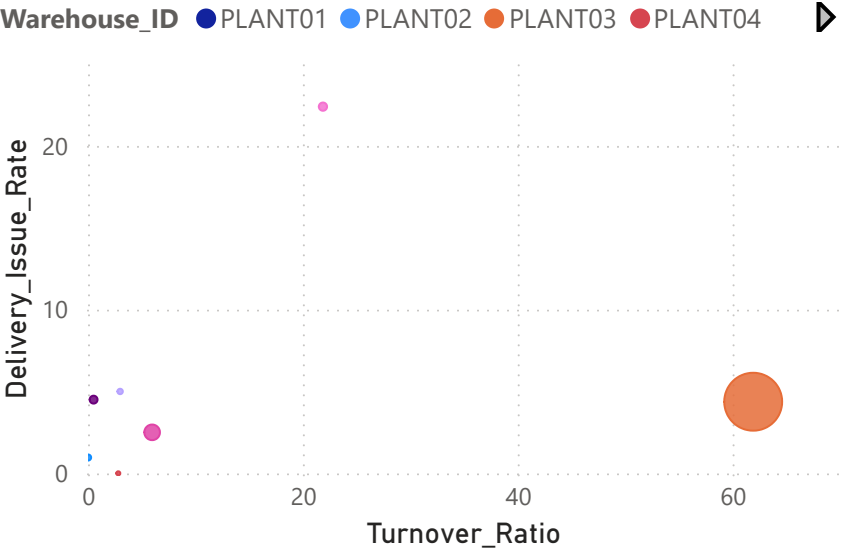
Insights: From this graph we can see category item is available in how many warehouse.
 There is only one category so this category is available in all warehouse. Total count is 19.

Shipped_Inventory_Ratio by Month and Warehouse_ID



Insights : Operational dependency risk: Heavy reliance on PLANT03 could be risky if disruptions occur. Diversifying shipment loads may improve resilience.

Turnover_Ratio, Delivery_Issue_Rate and Total_Shipments by Warehouse_ID



Insights : High turnover warehouses slightly increase delivery issues, but PLANT03 maintains a reasonable balance compared to others.

946.43K

Total Logistics Cost

921.53K

Total Shipment Cost

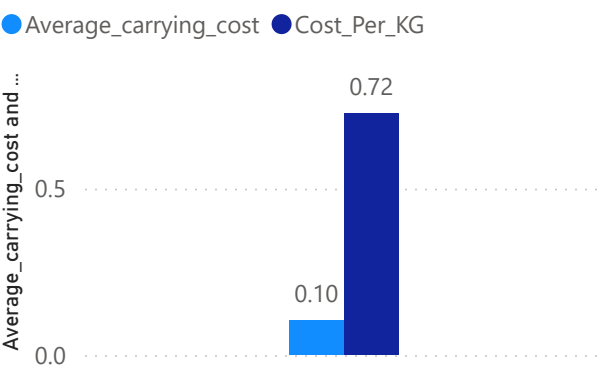
11.23K

Carrying Cost Total

Carrier_Name	V44_3		V444_0		V444_1		Total	
Warehouse_ID	Cost_Per_Shipment	Cost_Per_KG	Cost_Per_Shipment	Cost_Per_KG	Cost_Per_Shipment	Cost_Per_KG	Cost_Per_Shipment	Cost_Per_KG
PLANT03	10.27	1.32	1.76	0.21	9.76	1.07	6.75	0.76
PLANT04	10.27	4.89					10.27	4.89
PLANT08	10.27	2.05	1.73	0.12			1.75	0.12
PLANT09			2.20	0.13			2.20	0.13
PLANT12			1.98	0.20	9.78	0.56	7.79	0.50
PLANT13			2.75	0.17			2.75	0.17
PLANT16	10.27	0.36	22.70	1.50			14.35	0.60
Total	10.27	0.92	1.81	0.21	9.76	1.02	6.75	0.72

Insights : For the main carrier–warehouse pairs, cost per shipment and cost per kg vary noticeably, with some plants. PLANT04, PLANT16 showing much higher cost-per-kg than others.

Average_carrying_cost and Cost_Per_KG



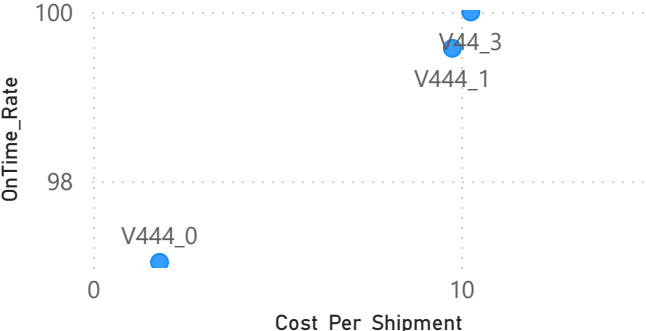
Insights : Average carrying cost is small compared with transportation cost per kg, confirming that inventory holding is a secondary cost driver on this route.

Cost_Per_OnTime_Shipment by Warehouse_ID



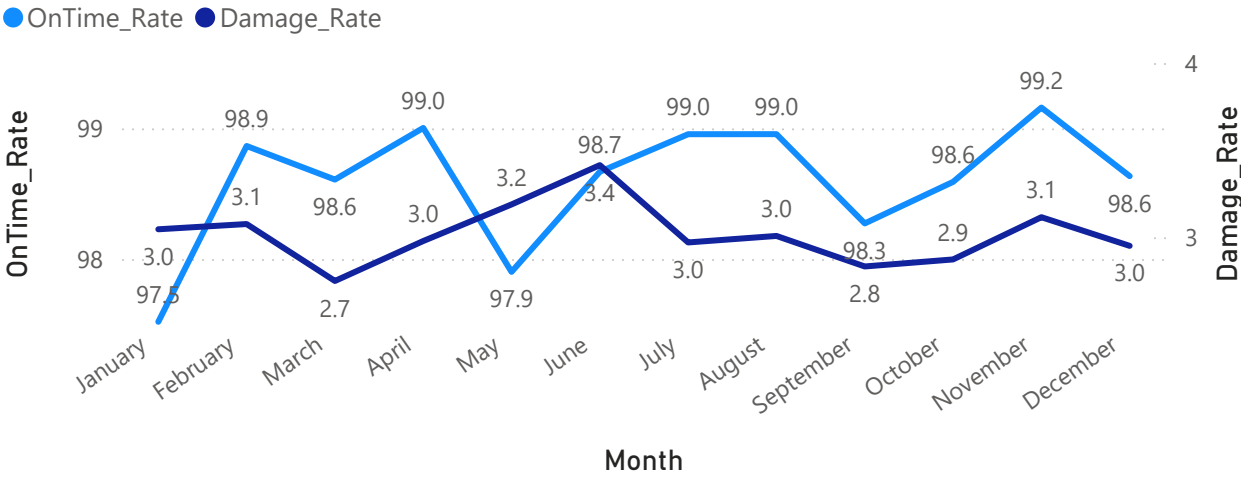
Insights : A few warehouses have significantly higher cost per on-time shipment than others, indicating inefficient combinations of cost and service from those locations.

Cost_Per_Shipment and OnTime_Rate by Carrier_Name



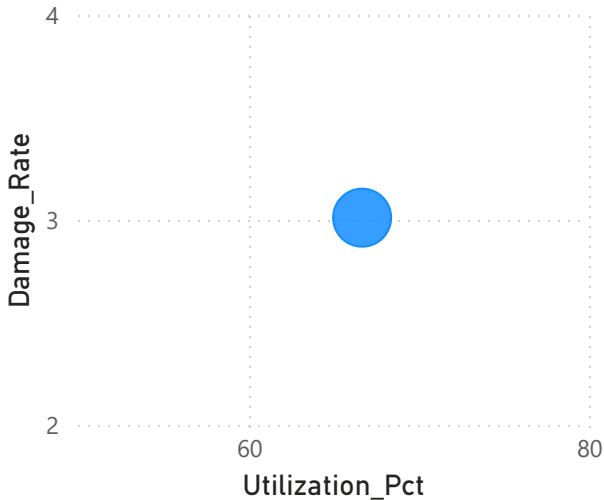
Insights : One carrier achieves a high on-time rate at a moderate cost per shipment, while another carrier is both more expensive and slightly less reliable

OnTime_Rate and Damage_Rate by Month



Insights : Here we see the trend for on time rate and damage date by month.
we can see least on time rate is 97.5 and highest on time rate is 99 and least damage rate is 2.7 and highest damage rate is 3.4.

Utilization_Pct, Damage_Rate and Total_Shipments



Insights : The plotted warehouse/plant operates in a mid-to-high utilization band while keeping damage rate moderate, suggesting that current workload is not yet creating severe quality stress

Destination_Region	PORT09		Total	
Carrier_Name	Vendor_Risk_Level	Delivery_Issue_Rate	Vendor_Risk_Level	Delivery_Issue_Rate
V44_3	LOW	2.46	LOW	2.46
V444_0	LOW	7.62	LOW	7.62
V444_1	LOW	2.42	LOW	2.42
Total	LOW	4.40	LOW	4.40

Insights : All carriers serving this destination are classified as LOW vendor risk, meaning none of them currently trigger concern on combined late and damage performance.

Warehouse_ID	PORT09	Total
		0.00
PLANT01		0.00
PLANT02		0.00
PLANT03	1.75	1.75
PLANT04	0.00	0.00
PLANT05		0.00
PLANT06		0.00
PLANT07		0.00
PLANT08	8.96	8.96
PLANT09	2.00	2.00
PLANT10		0.00
PLANT11		0.00
PLANT12	1.00	1.00
PLANT13	1.80	1.80
PLANT14		0.00
PLANT15		0.00
PLANT16	0.39	0.39
PLANT17		0.00
PLANT18		0.00
Total	1.76	1.76

Insights : Only a few warehouses (e.g., specific PLANT IDs) are contributing shipments for this port, while many others show zero volume, indicating a concentrated routing pattern.

